Consensus sequence – An ideal sequence in which each position represents the base which is found most often.

In addition to General transcription factors (GTF) and **RNA Pol II, a mediator** is required for transcription. The interactions between RNA polymerase II and regulatory TF that bind to enhancers or silencers are mediated by a mediator.

RNA Polymerases cannot bind directly to the DNA, first it binds to the transcription factor which recognizes the promoter sequences which helps to find the protein coding regions of DNA.

RNA Polymerase with the promoter sequence will transcribe the gene. Transcription factor plays an important role in guiding RNA Polymerases to the promoter sequence. RNA Polymerases bind RNA nucleotides together forming a growing strand in the 5' → 3' direction. Transcription occurs in 5' → 3' direction, RNA Polymerase catalyses the addition of nucleotides at the 3' end of the growing chain of RNA.

In Eukaryotes, 3 different RNA Polymerases called RNA Polymerases I, II and III are found.

Enzyme	Synthesis
RNA Polymerase I	Large Ribosome RNAs except 5S rRNA
RNA Polymerase II	Precursors of mRNAs (hnRNAs)
RNA Polymerase III	tRNAs, 5S ribosomal RNA, snRNAs (small nuclear RNA)

The processing of pre-mRNA to mature mRNA / Molecular mechanism of RNA modification

In eukaryotes three major types of RNA, mRNA, tRNA and rRNA are produced from a precursor RNA molecule termed as the primary transcript or preRNA. The RNA polymerase II transcribes the precursor of mRNA, which are also called the **heterogenous nuclear RNA** or hnRNA which are processed in the nucleus before they are transported into the cytoplasm.

Capping

Modification at the 5' end of the primary RNA transcript (hn RNA) with methylguanosine triphosphate is called capping.



Internal methylation
Apart from capping, the internal nucleotides in mRNA are also methylated. Methylated sites are present in translated, untranslated regions, introns and exons.

Purpose of Capping

1. Protects RNA from degradation.
2. Capping plays an important role in removal of first intron in pre mRNA.
3. It regulates the mRNA export from the nucleus into the cytoplasm.
4. It helps in binding of mRNA to the ribosome.

Tailing / Polyadenylation

The 3' end of hnRNA is cleaved by an endonuclease and a string of adenine nucleotides is added to the 3' end of hnRNA (pre mRNA) is known as Poly (A) tail - Polyadenylation. This process is called tailing or polyadenylation.

Purpose of Tailing

1. Translation of RNA transcript is facilitated.
2. Helps in the synthesis of Polypeptides.
3. It enhances the mRNA stability in the cytoplasm.

The protein coding regions are not continuous in eukaryotes. Split genes were independently discovered by Richard J Roberts and Phillip A. Sharp in 1977 and was awarded Nobel Prize in 1993. **Exons** are the coding sequences

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